

Numerical Calibration of Premium under Fixed Benefit Constraints in Life Insurance Models

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Overview

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- Calibration problem in life insurance product modeling
- Target: prescribed maturity value
- Constraint: fixed benefit structure (sums insured)
- Unknown: premium level consistent with target outcome

Core Modeling Problem

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- Product valuation defines mapping f of premium P onto maturity value MV :

$$MV = f(P)$$

- Goal: for given MV_{target} to find P^* satisfying

$$f(P^*) = MV_{\text{target}}$$

- No closed-form expression for f
- Each evaluation requires full policy projection

Mathematical Structure

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- Root-finding problem:

$$f(P) - MV_{\text{target}} = 0$$

- Function characteristics:
 - numerically evaluated
 - driven by multi-period cashflow projection
 - typically monotone in P
- No derivative available

Decomposition Logic

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- Premium affects:
 - savings / investment accumulation
 - cost coverage (risk, expenses)
- Maturity value arises from:
 - accumulated account value
 - reduced by charges and risk costs
- Implicit coupling across time:
 - premium $\rightarrow$ cashflows $\rightarrow$ account value $\rightarrow$ MV

Numerical Method

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- Iterative search over premium P
- Initial approach:
 - bisection on bounded interval
- Enhanced approach:
 - adaptive interval updates (moving boundaries)
 - faster convergence
- Typical convergence:
 - ~ 15 iterations
 - precision: currency level (2 decimals)

Implementation Aspects

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- Each iteration:
 - full projection of policy cashflows
 - computation of maturity value MV
- Key considerations:
 - computational cost per evaluation
 - numerical stability of convergence
 - robust stopping criteria

What was actually solved

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- Transition from forward valuation to inverse calibration
- Stable numerical procedure for premium determination
- Integration into actuarial projection framework
- Consistent handling of modified cost structures

Outcome

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- Reliable computation of premium for target maturity value
- Reduced iteration count via adaptive method
- Numerically stable within production constraints
- Applicable to broader product configurations